



E-COMMERCE

Mobile Application

Submitted By

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Abstract

The 21st century is the era of increasing use of technologies in business leading to e-commerce applications. Business organizations offer mobile application technologies that have flexible and agile-based structures to respond to customers and business conditions. Electronic commerce is a way of exchanging products and services over the Internet. E-commerce enables organizations to have virtual operations for their daily activities. E-commerce applications lead to lower expenses, access to complete markets, and attain new customers. E-commerce applications have also been increasing worldwide at a higher level. This document aims to focus on the e-commerce application structure with emphasis on the importance of e-commerce applications for business organizations. Every business at the national or international level trying to increase its demand by adopting technology-based activities. E-commerce helps decrease the average loss rate of companies as well as countries. The development of this application is to inculcate modern technology in business. This application has been converted from manual commerce to e-commerce. It offers services worldwide by responding to customers. The development of this application is to overcome the issues of commerce. The idea seems simple, but the development of a customer fascinating application is a little bit complicated. According to the requirement of the business, the application has four panels having different activities.

Chapter No 1

INTRODUCTION

1.1. Introduction

The growing use of technology and modern devices has been changed the customer and seller atmosphere. Travelling in the market may cause some complications. Customer can deal with another daily routine in the time which they utilize for travelling. This application will create ease of purchase of products for the users by installing and running this application on any device supporting installation. Customer will have the best experience after using the developed application. The project aims to provide a more comprehensive- range of product sale and purchase. Customers can look at each product and give their review about the product after purchasing from the app. The developed application provides multiple interfaces by securing customer personal data from any breach. Customer financial information is most important for the customer. By using this application, customer can do transaction of money during check out. Customers can just login into their account deals with products and all other services. With the increasing use of technology and devices, e-commerce is becoming popular among customers and sellers. All the business platforms are shifted to online services. By increasing the online sale and purchase system, this app is developed and fulfil all the customers and sellers' current requirements. People attracted more when they use technology in their daily lines. This application is a complete package in its nature which facilitate customers and sellers as well. This application creates an atmosphere of communication between both customers and product and services providers. The actual focus is on the developmnt of design and functionality of this mobile application is to attract consumers with a user-friendly interface(Jabbar et al., 2019).

1.2. Interfaces of Application

This application consists of the following interfaces:

- **Admin Panel**

One interface is for admin from where the admin can control all the activities. Admin can deal with products, customers and even with resellers. Part of admin is central for controlling the whole system. That can prove as intermediate handling for this interaction.

- **User Panel**

This is an interface is for customers. Attractive GUI bases interface will provide an easy-to-use platform where an illiterate person can operate and make a purchase of products. A comprehensive and self-explanatory UI increases the useability of the application.

- **Reseller Interface**

This is an interface is for reseller from where they can purchase products and manage their orders. Reseller can also control the stocked products. All the activities of the reseller managed. A reseller does not need to have the permission of the admin for making future shirt decisions.

- **Guest User**

This is an interface for guest users to access the products and view the details of the products. They can also perform limited activities by using this e-commerce application. The functionality is a little bit like the registered user interface. Guest user can make the ideas of the products by searching on the pages.

By the development of complete application in the form of these panels, every person can provide the facility of their services in the right way. Each interface has its emphasis on the working of the application (Jagtap & Hanchate, 2008).

1.3. Client Introduction

Name: Pratibha

Company: 07 Services

Website: www.o7services.com/

Email:

1.4. Project Background

Interfaces of the application are designed to increase the customer interest for using application and trust for e-commerce. Mobile commerce with its popularity also increasing the demand for e-commerce in the business market. When designing such applications the important thing when designing such applications is to design the applications in such a way that it does not distract the user and make them able to use application for the original purpose for which it has been developed. However, real concerns of the application should not be avoided that are related to the accessibility and securi. All All stored sensitive data like financial, or personal information

on mobile when designing such applications the sensitive data should be kept in the complete secure platform where it would be secured and free from unauthorized access. These measures should be applied before actual transaction. The actual vision of this application is to provide a secure and valuable commerce platform for customers (Gültekin & Bayat, 2014).

1.5. Problem Statement

The previous problem was that there was no virtual place exists for the activities of shopping at a central place. Each and everything was done manually. The company did not have sales reports and progress reports properly. The traditional way of record controlling activities was complex. This could create a situation where immediate decision or step difficult to take. Virtual data storage place can make the owner of the business able to access any record as and when required. Admin or owner cannot decide about several customers and business progress level without finding their records. They also did not have any proper resource for the comparison of products and sale (Åkesson, n.d.).

The customer also felt difficulty physically approaching the products in the market. They had to spend more money and time shopping. They also did not have a range of products with proper description from those they could select the required one. Customers were not able to know about trending products and brands because the market is not a place where trending statistics based on current fashion or user choices. It can be possible only when users buy things from a virtual or central place that all the activities are recorded in the system. Customer set product trend when they constantly search for the same product according to the current requirements (Ivanov, 2012).

1.6. Clients Requirements

1.6.1 Requirements

The client's requirements includes building of four interfaces for the users. These can be the admin interface, reseller interface, customer interface, and guest user interface. Each one is facilitated by using his interface. Every stakeholder can access the system and perform his task. There must be security measures applied to the system. For example, only admin can add or remove data from the application and guest user can only view products and he cannot buy until register in this mobile application.

1.6.2 Functional Requirements

The main functional requirements specified for the projects are:

Table 1: Functional Requirements

<i>US No</i>	<i>User Story</i>
<i>R_1</i>	An admin can manage the whole application.
<i>R_2</i>	An admin can log in.
<i>R_3</i>	An admin can edit my profile to make changes to my details.
<i>R_4</i>	An admin can view products.
<i>R_5</i>	An admin can add products.
<i>R_6</i>	An admin can modify products.
<i>R_7</i>	An admin can view categories.
<i>R_8</i>	An admin can modify categories.
<i>R_9</i>	An admin can view the overall sales of products.
<i>R_10</i>	An admin can view user profiles.
<i>R_11</i>	An admin can edit users of the application.
<i>R_12</i>	As an admin can view resellers' progress.
<i>R_13</i>	An admin can view and edit the profiles of resellers.
<i>R_14</i>	A user can log in with an email and password.
<i>R_15</i>	A user can view my profile.
<i>R_16</i>	A user can view the profile and change the password.

<i>R_17</i>	A user can view order details.
<i>R_18</i>	A user can check my wallet balance.
<i>R_19</i>	A reseller can register my account.
<i>R_20</i>	A reseller can log in.
<i>R_21</i>	A reseller can view the stock of the product.
<i>R_22</i>	A reseller can view all the available categories one for my purchases.
<i>R_23</i>	A reseller can add to the cart all the products.
<i>R_24</i>	A reseller can get a discount.
<i>R_25</i>	A reseller can purchase products.
<i>R_26</i>	A reseller can view products details for me.
<i>R_27</i>	A guest user can view the available categories.
<i>R_28</i>	A guest user can view the products.
<i>R_29</i>	A guest user can view the products in 3D form.
<i>R_30</i>	A guest user can view products details.
<i>R_31</i>	A guest user can add products to the cart..
<i>R_32</i>	An admin can manage the whole application.

1.6.3 Non-Functional Requirements

These all are mentioned below:

- The admin of the application should be able to manage and maintain the security checks.
- It is mandatory to apply constraint on the application so it would be more secure, reliable and performnall the jobs specified in the requiremwnt
- There is a need of development of user friendly interfaces.

- Users should easily access the application for use.
- The transaction should be secured and error-free.
- The application should be free from instabilities.
- Shopping should be reliable and attractive.
- The authentication process must be strict and functional.

1.7 Stakeholders

- **Reseller**

A reseller is an individual that resells the services and goods of other sellers instead of the consumption of goods like other customers.

- **Customers**

Customers are the actual people who purchase goods from the resellers or the producers. They can view the products and services and purchase these.

- **Admin**

Admins control all the data on the website. Website management is the main activity that the admin performs from the admin panel. He can feed the records in the database.

- **Guest user**

Guest users are customers but without login credentials. They can do almost everything a customer can but cannot checkout or do payments.

- **Development Team**

The development team is answerable for the development of the complete mobile application. The team is also accountable for the proper functionality of the application throughout its use.

Chapter No 2

PROJECT OBJECTIVES

2.1. Objectives

The objectives of this application are:

- The application has the objective to facilitate the buyer for online shopping.
- It has the objective to make shopping easy and time-saving.
- It has the objective of providing up to date detail of the product.

- It has the objective of reducing overall management expense.
- It has the objective of increasing customer trust in the supplier.
- It has the objective of making services related to e-commerce more efficient.
- It has the objective of increasing the monthly sale of the products.
- It has the objective of increasing the number of customers by providing them on time and original products.
- It has the objective of making a responsive platform for e-commerce activities.
- It has the objective of providing a range of products on remote access devices.
- It has the objective of enhancing the business of the client.

2.2 Project Aim

The project aims to provide the customer with a platform for electronic commerce. With the advancement of technology, everything is transforming into a digital environment. This project is developed so that customer can view and buy products from home using remote control devices. The application can offer a variety of categories and products to the customer. Admin can view the overall sale of the system without processing manual reports. Admin can also view the progress of the reseller by just clicking on the button, the system automatically will display the overall progress of each reseller. Reseller and customers can view the category and buy the required products. Reseller can also get discount on the products. Guest users can visit the application and view the products.

Chapter No 3

ACADEMIC RESEARCH

3.1 Literature Review

Mobile commerce or E-commerce is a quite an updated image of the digital business that entered the business vocabulary during the time span of 1970s. Emergence of a type of electronic commerce in which internet has become an essential dial tone of the for running the business in the modern era. The author discussed the role of electronic commerce and the present and future impacts of electronic commerce. There is also demonstration of interactions between business

and customers, and the evolution of indecision. Electronic commerce definition and typography are offered (Wigand, 1997).

The World Wide Web has not changed the phenomena of responding to the users and behaviour of people around the world. Because of this the emergence of electronic commerce was possible in the business market which has affected the lives of general public. Many other countries also started online shopping of products. People are still not accustomed to shopping through digital channels but with the passage of time gradually consumers moving towards the changed nature of shopping (Rahman et al., 2018). Different people view e-commerce from different perspectives. The current e-commerce framework includes the specific and rigid core factors of e-commerce. The new situations and challenges are constantly faced by the market of e-commerce. To address these complicated situations, a functional e-commerce framework is being introduced. This framework consists of six major e-commerce components, each containing several examples (Poong et al., 2006).

E-commerce involvement is growing rapidly, with the development of internet technology. The electronics contract is an important part of the electronic commerce platform. The thing that provide a guarantee of the rights of members and increase the trust of customers is the agreement between participating parties. Contracts formed by using online resources can be e-contracts or TPAs of agreement between all trading partners trading partners e.g. stakeholders. This agreement also included payment options, buyers, and sellers (J. M. Authors: Biswajit Tripathy, 2018).

Electronic commerce is customizing many aspects of social life and business. New strategies are required in the environment of business. It enhance the power of the competition in the market. E-commerce is a roadmap of running new businesses. Its impacts are growing every year (Rahayu & Day, 2015). The rapid spread of the internet facilities caused rise in the activities of e-commerce. It is now transformed into a common trend in the today's world. Information groups and web-oriented economical models are a new business reality, as organizations and individuals find it easier to purchase goods and services off the coast. However, most of the developing countries are far from experiencing the fact that many factors hinder the growth of e-commerce (Alyoubi, 2015).

As users are working on the internet and increasing gradually, different strategists are used for the internet to create a competitive environment. It has transformed the actual trading field by

helping the brands to create their online serving channels. The firms or industries that are newly established and introduced e-commerce sites have competed with others to create brand awareness in the digital markets through e-commerce. They are able to drag their business facilities online through the help of these strategies and replace those benefits. It can be done through e-commerce initiatives to establish a new brand in the competitive environment (Mohapatra, 2000). Impressed with the intensity of product information and the CEO's innovation in adopting websites. The key factors suggested the oddity of the adoption trend in SMEs. However, the factors that were shown to be important and the suggestions arising from these factors brought to the conclusion which provide the likelihood of the weakness of the trend of acquisition of EC in SMEs in NZ (Al-Qirim, 2007). (Štefko et al., 2019) incited that the analysis of the use of digital applications reflects the influence of digitization as the emergence of upgraded communication platforms is altering consumer shopping practices in the digital and remotely access environment. The study of trust component has been carried out in articles from business to psychology. The use of mobile phones for future demand attracts customers (Keng Siau & Shen, 2003).

There are special benefits of mobile services investigated by (Nilashi et al., 2015), resolving trust issues is a major concern in the implementation of mobile commerce. From vendors, providing customer confidence is a complex process and relatively difficult to achieve. An important step in gaining customer trust and attracting them to mobile commerce is building trust Websites for mobile software and transactions. . (Zhang Jonathan Lazar Scott McCoy Program Chair & Hess Jeff Kim, n.d.) formulated the improvements with mobile-based consumer personality and advertising content terms. Despite growing market access, however, some pop out of trends divide the web-driven marketplace into untold locations. Markets as networks and internet-based distributed technologies enhanced the wau of conducting business. The killer applications for customers will be the same that allows mixing and matching of products and services in private and in real-time (Whinston, n.d.).

Interest in the concept of "interactivity" has grown as we enter a "perpetual" society where People can talk anytime, anywhere. Despite the importance of communication in the mobile commerce (MC) Environment, a topic that has acquired little attention in the academic perspective. The effects of interactivity components and interactivity feedback on customer confidence and Intent of transaction in MC are assisting in the future growth of the business.

Experimental results show the addition of specific components of the interaction KMC improves the model (Lee, 2005). As service-oriented computing grows, so do e-contracts to help automate business partner agreements and contracts. The key challenge is to introduce traditional contracts into EE contracts that facilitate runtime monitoring and management (Krishna & Karlapalem, 2008).

It is represented by (K Siau et al., 2004) significant growth in mobile commerce e-commerce offers customers to access, dominance, mobility, and localization. Trust is a considerable hurdle to adaptation and development despite the capacity of mobile commerce. Many consumers feel uneasy about this idea trade on wireless, handheld devices. There is a framework of trust developed from a network of central objectives. Advancement in wireless technology has accelerated advancement in electronic commerce Use of mobile devices(Zhang Jonathan Lazar Scott McCoy Program Chair & Hess Jeff Kim, n.d.). There are several benefits to using e-commerce. The electronic solution will reach as many people as possible. At present, there is very little effort to accomplish anything. In this environment of digital access to reach the thousands of customer at a time is possible and it will cost only a little amount of whole revenue of what it is today. The employment of mobile or internet- based commerce introduced a new world of commerce that was not possible in the past, as well as in the industrial framework, the main obstacle that is not the use of the early internet is present everywhere (Anumba & Ruikar, n.d.).

3.2 Case Studies

3.2.1 Case Study: 1- kalahari.net

It is a South African oriented business mentioned as an e-teller by its owners and managers. It is a web- based digital facility. Kalhri.net is an electoronic retailer and business model in the business to consumer sector focused on individual customer sales. Sean Peter, Dan Remini, and Andre de la Harpe are businesses that sell books, videos, CDs, DVDs, videos, software utilities, hardware components, ware, alcohol, and healthcare equipment. These products are brought from different areas of South Africa, and elsewhere. Moreover, to these products, kalahari.net has several online partners those assist to purchase services such as theatres, cinema ticketing solutions, and major events. It was the mistake of the firm misunderstood the market plans. The technical complications and challenges, especially those related to information system

architecture, were not recognized. The website interface is explained as being "horrible" with time consuming download feature, and lack of proper information on the website. There was unreliable data on the web and the process of loading data consumed days or weeks. There was no presence of completion process cycle on site. The team of kalahari.net consisted of three business officials, and about ten new graduates did not think well through IT students had a sourcing policy. There was no proper plan of fund receiving through kalahari.net, as the basis for the "open book" for the financial support they had. In addition, there were insufficient official controls to manage the business this led to the failure and distraction. The loss of potential amount was not large but the failure of electronic commerce was on the surface in the full light of public media. This was one of the most unexpected reason of chaos.

3.2.1.1 Findings

In the above case study it is cleared that a business with e-commerce strategies can lead to initial chaos if proper solutions to the decline are not adopted. With the start of new technology without proper management, it is difficult to grow business steadily. With e-commerce application, advanced features are required to deploy on the web. Time management and accuracy are most important while searching through the web.

3.2.1.3 Providing solution to the Problem

The best resolution of the raised complication is to improve the user's operational search feature, usually to signal the beginning of a potential transaction. As a result, the search feature should be constantly checked. Reports should come to the marketing places to examine the interests of the buyers. Second, the loading time for web pages is considered important. The loading time should also be monitored from time to time to improve the time management facility of the application.

3.2.1.2 Conclusion

It is concluded that the current structure of IS function is performed at a satisfactory level and as a result minimum error rate is recorded in the database of products. Turn payment methods into feasible gateways and make checkout processes easy. There are many options for the payment, all of which functional properly. It is also a powerful part of the relationship with the relevant 3rd parties as the website was voted by the public as the best 1000 sites. This is an evidence that it is about fulfilling the needs of the client.

3.2.2 Case study: 2-Skullcandy

Skullcandy was turned out into a complication that required to be solved. The founder of the company rode a chairlift in the city park. He was enjoying music during riding on the chair but during this he had to make a call. Indeed, the technology did not allow him to listen call during listening music. So, he decided that it is the time of introducing new thing with the solution of this problem. That was the time of emergence of technology in the field of headphone business. He decided that he would make a company that would introduce new technology based and enhanced head phone in the market that would fulfill all the needs of the users. It would also fit the modern lifestyle. The moment was the changing moment which made the Skullcandy a powerful firm that respond the technology based needs of the customers. It changes the audio lifestyle of the customers and never declined ever. After passing fifteen years of success, the brand still has deep roots in music and games. Its authenticity of brand make it different from all other brands and make it a powerful in the competitions of the businesses. Today, Skullcandy users are young and audacious audio user – the people who enjoying music, seeking new experiences and steadily finding amazing opportunities.

3.3.2.1 Findings

Their previous commerce platform, for sales purchase activities, was expensive to manage and maintain. It was not very active, and their team was working slowly as well as utilizing external resources of development which produced outcomes in high cost. They aware of to choose the right structure and area for work and have to observe different functional areas options from every angle. As they go through the process of finding a new e-commerce atmosphere, they also evaluated other service providers. Magento was quickly rejected because they were on that platform before the SFCC and they faced lack of experience. Use of Shopify Plus was their second turn. After the span of three months they spent lot of time on multiple channels and technologies based platforms, they solidified their decision between Big Commerce and Shopify Plus. They realized that effective international e-commerce is the need of the time. This would provide the time and features they needed. They would not acquire much concentration of complete team. This is what they experienced in the sales force commerce cloud that was discouraging.

3.2.2.2 Providing solution to the Problem

Their platform relocation was quite challenging and a complex project with many dynamic components. They adopted many currencies, changed websites and modified languages. They provide many facilities live on the platform. Their relocation continued smoothly instead of encouraging launching structure. Their website looked beautiful after alteration in it and their revenue streams looked uninterrupted from any flaw. During the first month of start the improvement rate was observed gradually increasing and they won best looking and functional website design award. This progress set a wayforward for the firm.

3.2.2.3 Conclusion

In Big Commerce, they are saving significant money that spent on the cost of establishing platform and overall development. More significantly, they have a highly agile environment that can change rapidly to meet the needs of their business. The SaaS component of Big Commerce is great because there is no need to be worried about hosting which allows them to concentrate on the business factors of the website. They see the Big Commerce platform as a key element of their e-commerce ecosystem, as their users come to their website to learn about their products, whether from different platforms like Amazon or Be from a retailer site. Therefore, the change according to the upgrading technology proved to be beneficial for them.

3.2.3 Case Study: 3-Albertsons

Albertson's use store based atmosphere and digital channels to provide customer a new experience and fascinating opportunities. In the era of 2000's it was the first store providing the facility of online products delivery. Albertson has always been at the front of the successful industrial firms when it comes to e-commerce. Today, as he looks at the grocery industry, Albertson has moved faster and easier. They want to motivate their customers for cooking and selecting goods for the cooking process by offering them a variety of shopping options, and, where possible, personalized, relevant content. Initiation with the current data, Albertson realized the lack of quality measures of results of search in terms of integrity and connectivity. It also contributed to the naturally high bounce rates that Albertson endured. They want to turn to millet by enhancing the customer experience and increasing the size of the basket.

3.2.3.1 Findings

Central to this change was the search for the site. It is concluded that fifty percent if the e-commerce is committed to searching process. They track an opportunity to find a product and

get a solution that would reduce the time of increasing the actual size of the basket. This phenomena will increase the conversion rates.

3.3.3.2 Providing solution to the Problem

They discovered a solution after choosing Bloomrich Search and Merchandising (BRSM). They have seen a huge improvement, which is more than a search box. Gives the ability to browse the entire catalog and experience on this ecommerce website. They can also observe from this statistic that customers can reach to the products and add them to the cart in less time with SMS. Since the implementation of BRSM the speed of constructing basked was increased by twentyfive percent.. By improving working platforms and improving their online application facilities they grew business gradually.

3.3.3.3 Conclusion

The company found an issue of low customer visit on their site and low purchase rate. But the solution was discovered by improving search mechanisms to increase people interest in using the online application of the Albertsons. They improved services by upgrading the mechanisms of their e-commerce platform. They wanted to change their company success rate.

Chapter No 4

METHODOLOGY

4.1 Agile Methodology

To develop the project, we have used the most popular agile method these days. This is a repetitive approach that is used for every development process. It uses small development cycles that divide tasks into smaller ways called sprints. This division of labour differs from other models of development. In an area where other project development approaches have failed to apply, a methodology is very helpful and efficient. Other models of the waterfall model, for example, behave in such a way that they do not support the incremental changes that are all set at the beginning of development. Smell recurrence or recurrence is possible after the onset of development. But the agile method, on the other hand, is an alternative to the solution in the sense that I accept any repetition or modification during development and help to correct the error during development. This gives the client a level of satisfaction as it provides an

opportunity to review the plan throughout life. It is a disciplined project management technique that allows for continuous inspection of the project during the development life cycle. It uses a Scrum framework. It is a repetitive development framework and is the most widely used. It depends on the character or the timebox (Maruping et al., 2009).

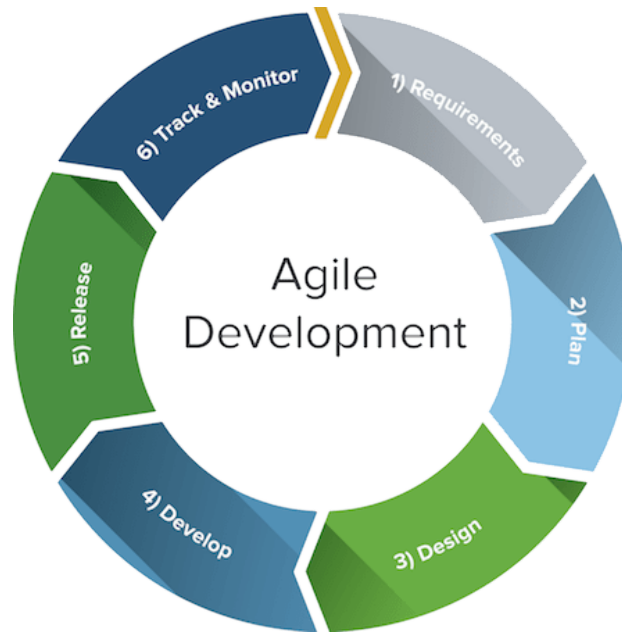


Figure 1: Agile Approach

4.1.2 Comparison with other Methodologies

Software development projects have long been a slow process, but Agile Methodology is aging. Software/systems have given rise to more than one adaptive approach Development. This is a technique for programming advancement that executes an iterative and precise way to deal with opportune programming conveyance. It is explicitly worked to adopt the advancing characteristics during the time of development. It adjusts the change to any scene, for example, client audits or adding any missing highlights. In the agile approach, the dangers are estimated over the existing pattern of programming improvement and the undertaking's turn of events. After iterations and changes, the product is making ready for delivery. After completion of each iteration functionality of the product is enhanced as compared to the non-iterative product. Every addition, in any case, increases new functionalities to the product. Agile or scrum methodology, the item should be conveyed within a brief period. It gives those functionalities to a speedy model called prototyping prior to make final delivery. In approach of agile, more importance is

given to working applications, associations, quick versatility, and client commitment rather than documentation. The waterfall strategy, then again, called a direct methodology(Thummadi et al., 2011). It does not react to evolving requests. In non-iterative systems, toward the beginning of the project, the determinations are resolved and settled. These conditions are unable to adjust after some stages. If a some of the standards has been missed out, we cannot return to the waterfall method. There is no specific limit of time under the non-iterative approach. It is feasible to deliver and develop the item in a sensible period. In the cascade strategy, more preference is awarded to documentation part. All the activities are precisely recorded. The strategy cycle creates a record that goes about as a contribution for the following stage of the waterfall (*The Waterfall Model and the Agile Methodologies: A Comparison by Project Characteristics*, n.d.).

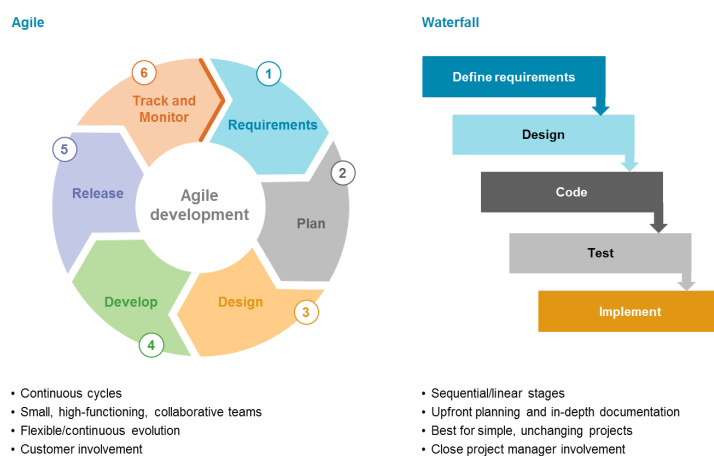


Figure 2: Comparison of Agile and Waterfall Approach

4.1.3 Approach for Project Development

The task development approach that we utilized is the agile methodology. We will utilize this methodology since it gives the way of making modifications. We will utilize the scrum method in this approach as our task will be carried out in 3 distinct runs. Through a light-footed procedure, the entire process is executed through various cycles. The significant benefit of scrum is adaptability as it accepts the progressions without any problem (Schwaber, n.d.). The task is to be inherent in 3 part. For each part, we will send it to the customer for criticism. The enhancements will be made in the run before beginning work on the following run dependent on that input. These three working parts are sprints.

That helps to divide the complicated task into sub-task and solve them one by one in an incremental way. Scrum has the following phases.

- Initiate the development process.
- Planning of the task for example we developed user stories after a complete analysis of the requirements. After the estimate, we developed a sprint backlog.
- The next phase is the implementation where execution of the tasks takes place to create the actual product.
- The review phase is to examine the whole work to make any modification or remove any flaw from the product development.
- The last phase is to deliver the final product of the sprint after reviewing or some changes.

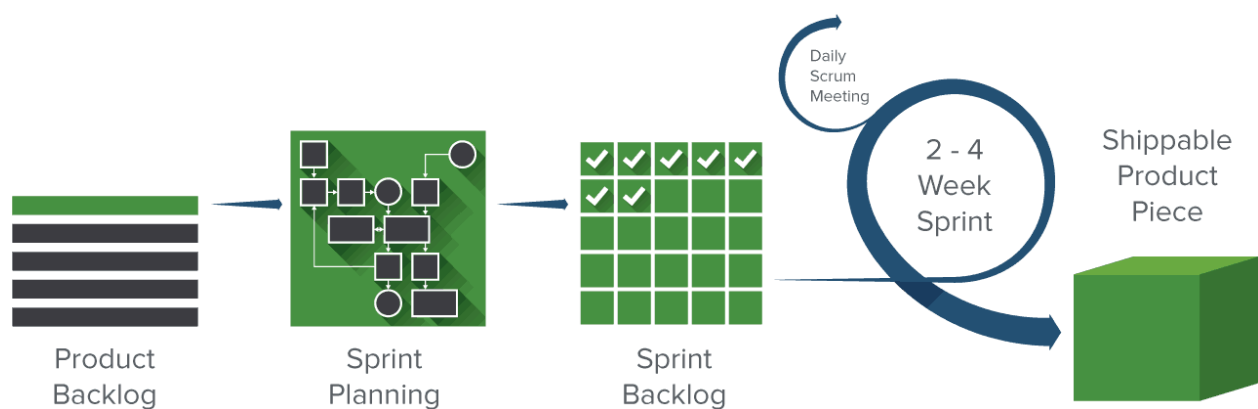


Figure 3: Phases of Scrum Based Agile Approach

Chapter No 5

PROJECT DISCUSSION

5.1 Description of Project

We have developed an e-commerce application for the client. It could be used to facilitate customers worldwide. The purpose of the application being developed is to facilitate customers. Saves the time and money which they invest in travelling from one place to another. Travelling in the market may cause some complications. Customer can deal with another daily routine in the

time which they utilize for travelling. This application will create ease of purchase of products for the users by installing and running this application on any device supporting installation. Customer will have the best experience after using the developed application.

The project aims to provide a more comprehensive- range of product sale and purchase. Customers can look at each product and give their review about the product after purchasing from the app. The developed application provides multiple interfaces by securing customer personal data from any breach. Customer financial information is most important for the customer. By using this application, customer can do transaction of money during check out. Customers can just login into their account deals with products and all other services.

With the increasing use of technology and devices, e-commerce is becoming popular among customers and sellers. All the business platforms are shifted to online services. By increasing the online sale and purchase system, this app is developed and fulfil all the customers and sellers' current requirements. People attracted more when they use technology in their daily lines. This application is a complete package in its nature which facilitate customers and sellers as well. This application creates an atmosphere of communication between both customers and product and services providers. The focus of the development of the application is to attract customers with a user-friendly interface.

5.2 User Stories and Related Screenshots

5.2.1 User Stories

The user stories of our project are given in the table below:

PB-No	User Story	Priority	Acceptance Criteria
US-1	As an admin, I want to manage the whole application so that I can handle the application.	Medium	Admin can manage the complete application.
US-2	As an admin, I want to log in so that I can manage the application.	Medium	Admin can log in to the application.

US-3	As an admin, I want to edit my profile to make changes to my details.	low	Admin can update his profile.
US-4	As an admin, I want to view products so that I can view all the available products.	Medium	Admin can view the overall products.
US-5	As an admin, I want to add products so that I can provide new products to users.	Medium	Admin can add products.
US-6	As an admin, I want to modify products so that I can update product detail.	High	Admin can edit the detail of the products.
US-7	As an admin, I want to view categories so that I can view available categories of products.	Medium	Admin can view categories of the products.
US-8	As an admin, I want to modify categories so that I can update product categories.	Medium	Admin can edit the categories of the products.
US-9	As an admin, I want to view the overall sales of products so that I can analyse how much profit the business gets.	Medium	Admin can view the overall sales.
US-10	As an admin, I want to view user profiles so that I can view all the users of the application.	Low	Admin can view the user's profile.
US-11	As an admin, I want to edit users of the application so that I can update the details of application users.	Low	Admin can edit users of the application.

US-12	As an admin, I want to view resellers' progress so that I can analyze how many sales they made.	Medium	Admin can view the progress of resellers.
US-13	As an admin, I want to view and edit profiles of resellers so that I can view and update their details.	Low	Admin can view and edit profiles of resellers.
US-14	AS a reseller, I want to register my account so that I can use the application.	Medium	The reseller can register his account.
US-15	As a reseller, I want to log in so that I can view all the products and my purchases	Low	Reseller can log in to the application.
US-16	As a reseller, I want to view the stock of the product so that I can purchase the products.	Medium	Reseller can view the product stock.
US-17	As a reseller, I want to view all the available categories so that I can select the suitable one for my purchases.	Medium	Reseller can view all the categories of the product.
US-18	As a reseller, I want to add to the cart all the products so that I can initiate the process of purchasing.	High	The reseller can add the products to the cart.
US-19	As a reseller, I want to get a discount so that I can get stocks at a discount.	High	Reseller can get discount on the products.
US-20	As a reseller, I want to purchase products so that I can use them for my reselling.	Medium	Reseller can purchase the stocked products.

US-21	As a reseller, I want to check the order detail so that I track back my orders.	Low	The reseller can check the order details.
US-22	As a reseller, I want to view product details so that I can select the best-suited products for me.	Medium	The reseller can view the product details.
US-23	As a customer, I want to log in with an email and password so that I can use the application.	Medium	The customer can log in to the application with his credentials.
US-24	As a customer, I want to view my profile so that I can check my details.	Medium	The customer can view his profile.
US-25	As a customer, I want to view the profile and change the password so that I can update my profile.	Low	The customer can change his password.
US-26	As a customer, I want to view order details so that I can check the progress of product shipment.	Medium	Customer can view all the products.
US-27	As a customer, I want to check my wallet balance so that I can make money transactions.	High	The customer can check his wallet balance.
US-28	As a guest user, I want to view the available categories so that I can select the required product.	Low	Guest user can view all the categories of products.
US-29	As a guest user, I want to view the products so that I can read all the detail.	Low	Guest user can view all the products.

US-30	As a guest user, I want to view the products in 3D form so that I can check product specifications.	High	Guest user can view the products in complete 3D form.
US-31	As a guest user, I want to view product details so that I can find the right one.	Medium	Guest user can view the product detail.
US-32	As a guest user, I want to add to the cart the products so that I can proceed to the next step.	Medium	Guest user can add products to the cart.

5.2 Screenshots

5.2.1 Admin Interface

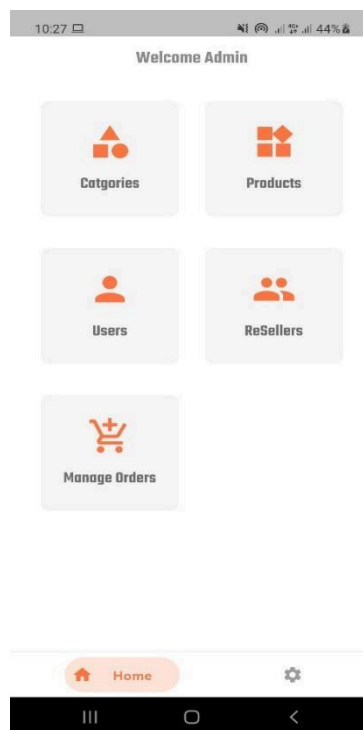
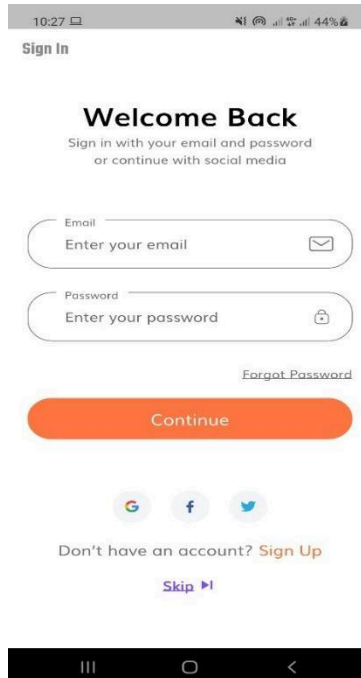


Figure 4: Main Page



10:27

Sign In

Welcome Back

Sign in with your email and password
or continue with social media

Email
Enter your email

Password
Enter your password

[Forgot Password](#)

Continue

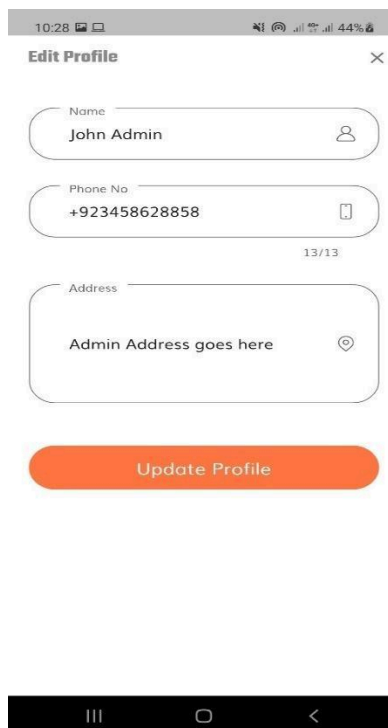
[G](#) [f](#) [t](#)

Don't have an account? [Sign Up](#)

[Skip](#)

III □ <

Figure 5: Admin login



10:28

Edit Profile

Name
John Admin

Phone No
+923458628858

13/13

Address
Admin Address goes here

Update Profile

III □ <

Figure 6: Edit Profile



Figure 7: View Profile



Figure 8: Manage Products

10:28 44%

← Add Product

Title
Enter product title T

Category
Enter product category

Description
Enter product description

Price
Enter product price

Stock
Enter product stock

No Image Selected Select Image

Add Product

Figure 9: Add Products

10:28 44%

← Update Product

Title
This is product T

Category
Category 1

Description
I will develop android app using flutter. Flutter is one of the popular development kit for

Price
12.5

Stock
10

No Image Selected Select Image

Update Product

Figure 10: Update Products

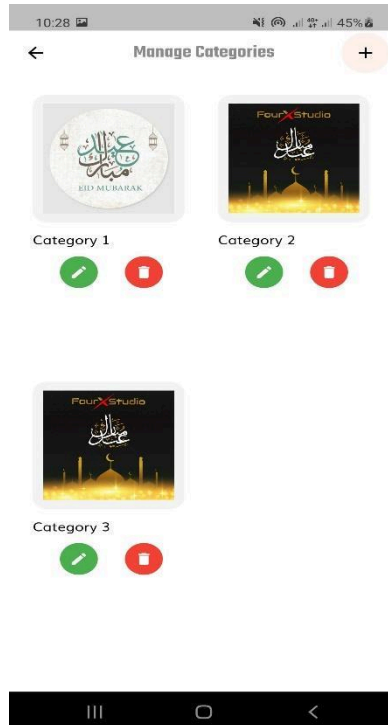


Figure 11: Manage categories

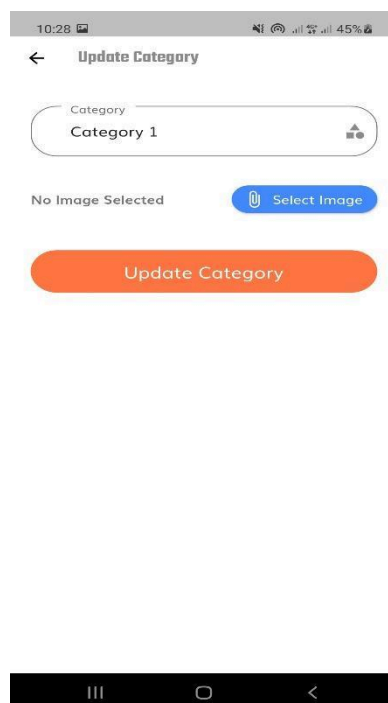


Figure 12: Update Categories

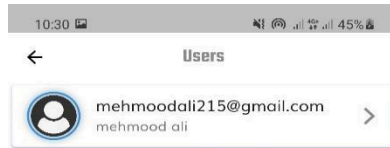


Figure 13: View Users

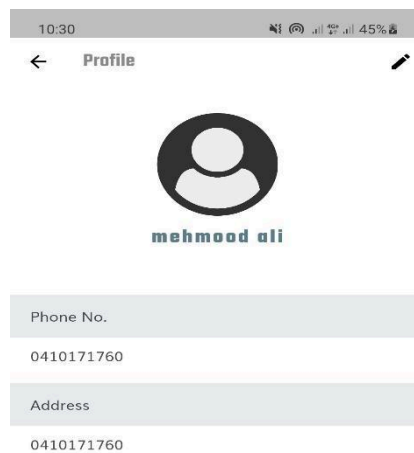


Figure 14: View User Profile

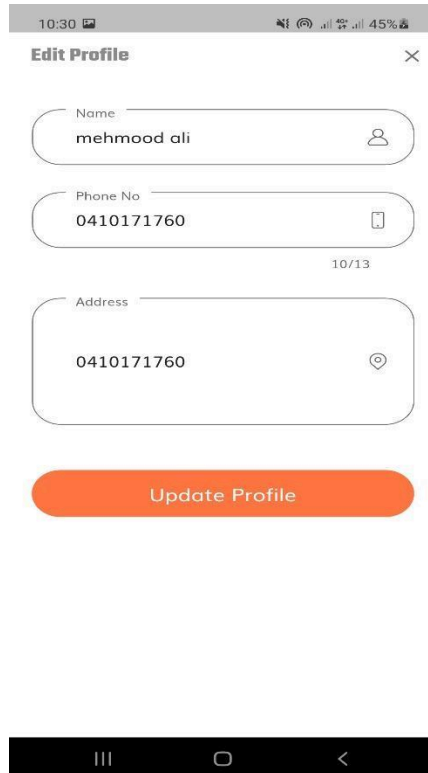


Figure 15: Edit user Profile

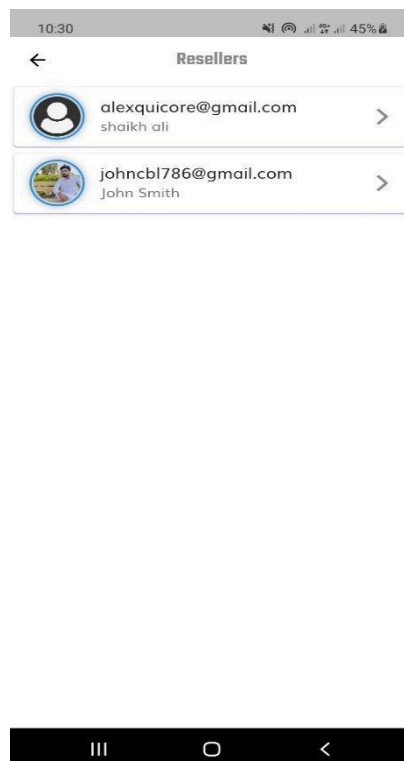
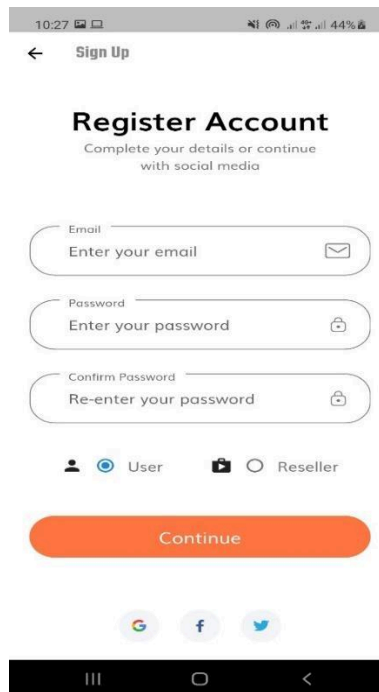


Figure 16: View resellers

5.2.2. Reseller interface



10:27 44%

← Sign Up

Register Account

Complete your details or continue with social media

Email
Enter your email

Password
Enter your password

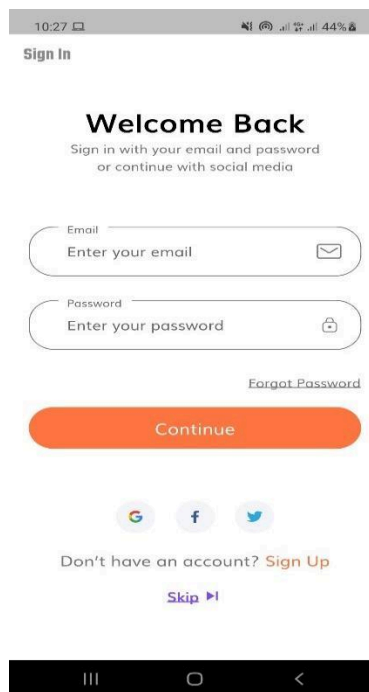
Confirm Password
Re-enter your password

☒ User ☐ Reseller

Continue

Figure 17: Reseller Registration



10:27 44%

Sign In

Welcome Back

Sign in with your email and password or continue with social media

Email
Enter your email

Password
Enter your password

[Forgot Password](#)

Continue

Don't have an account? [Sign Up](#)

[Skip](#)

Figure 18: Reseller login

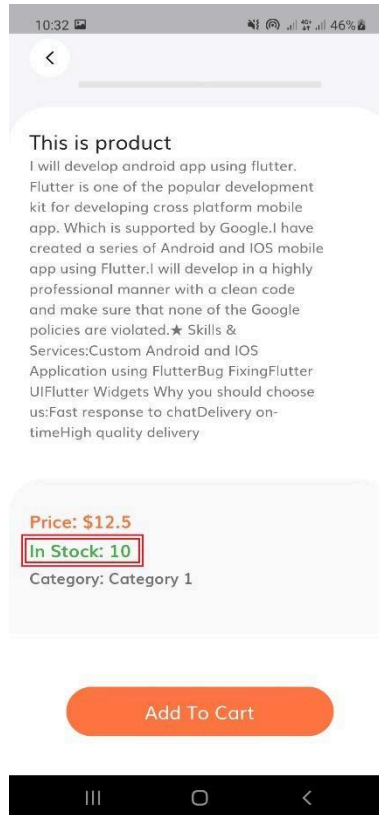


Figure 19: check Product Stock



Figure 20: view Categories

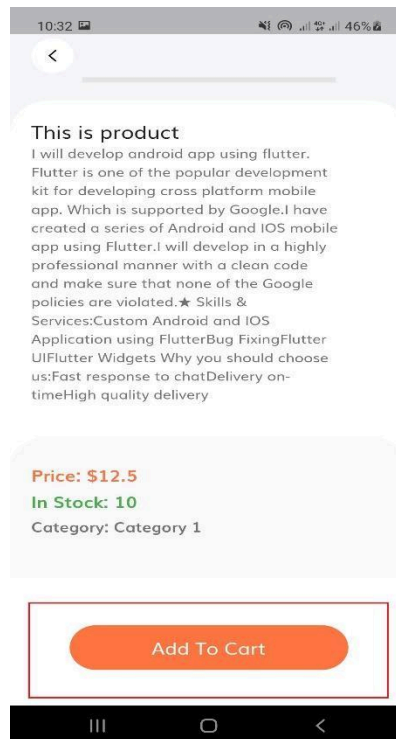


Figure 21: Add Product to Cart



Figure 22: Get Discount on Purchase

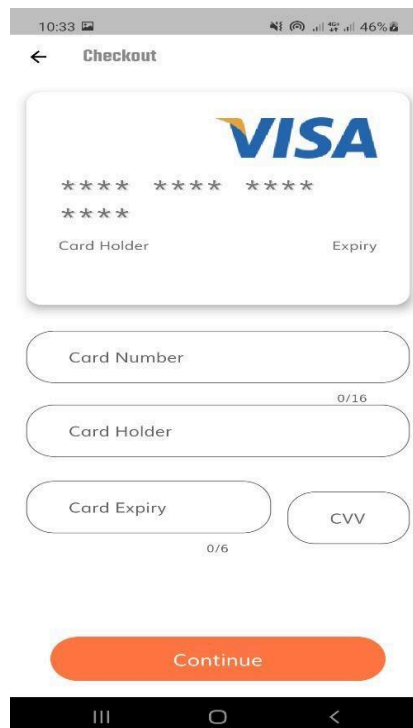


Figure 23: Checkout

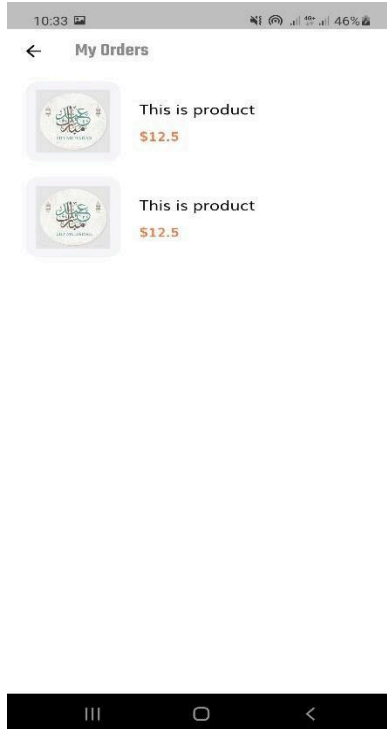


Figure 24: view Orders



Figure 25: track Order



Figure 26: view Products

5.2.3 Customer and User Interface

10:27

← Sign Up

Register Account

Complete your details or continue with social media

Email
Enter your email

Password
Enter your password

Confirm Password
Re-enter your password

☒ User ☐ Reseller

Continue

Google Facebook Twitter

Figure 27: Resisteration of Account for customer

10:27

Sign In

Welcome Back

Sign in with your email and password or continue with social media

Email
Enter your email

Password
Enter your password

[Forgot Password](#)

Continue

Google Facebook Twitter

Don't have an account? [Sign Up](#)

[Skip](#) ➔

Figure 28: login



Figure 29: View Profile.

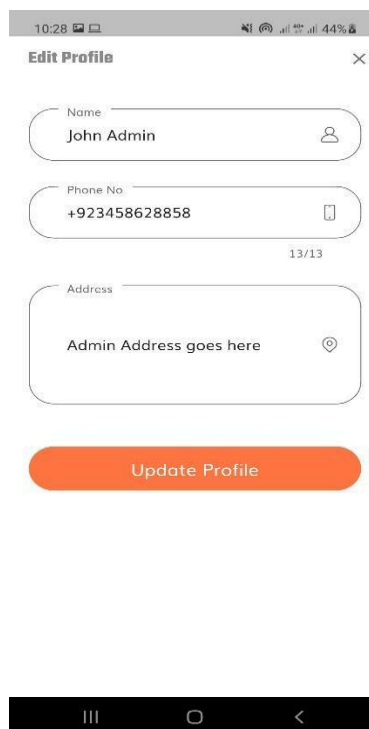


Figure 30: Update Profile

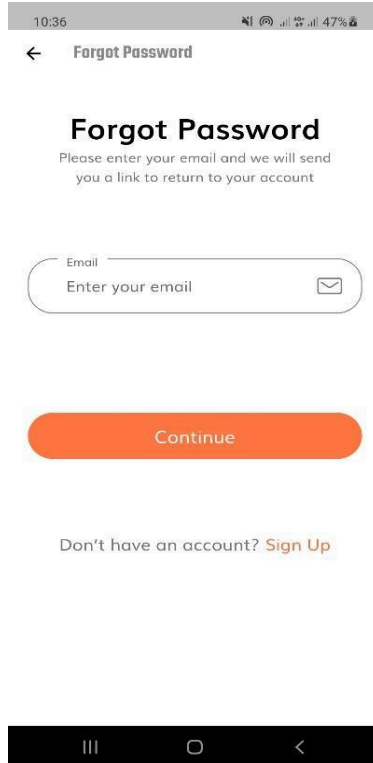


Figure 31: Update Password



Figure 32: View Orders



Figure 33Figure 34: View Products



Figure 35: View Categories



Figure 36: View Products

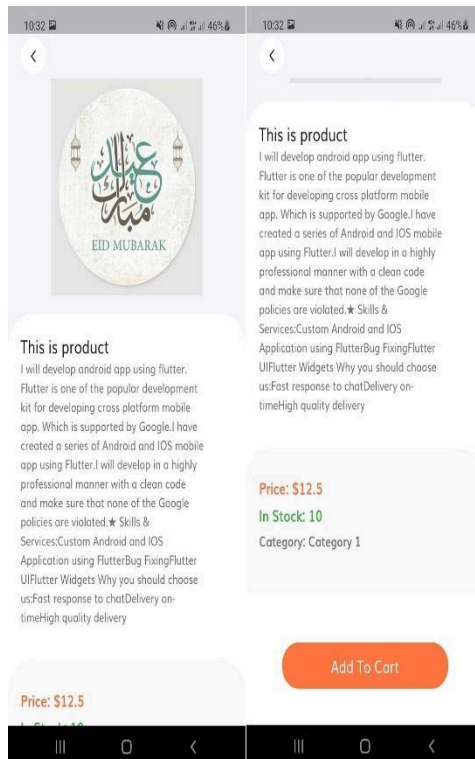


Figure 37 Product detail view Order

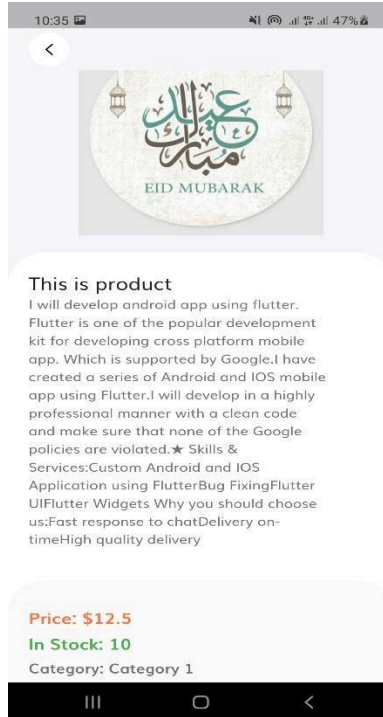


Figure 38: View Product Detail

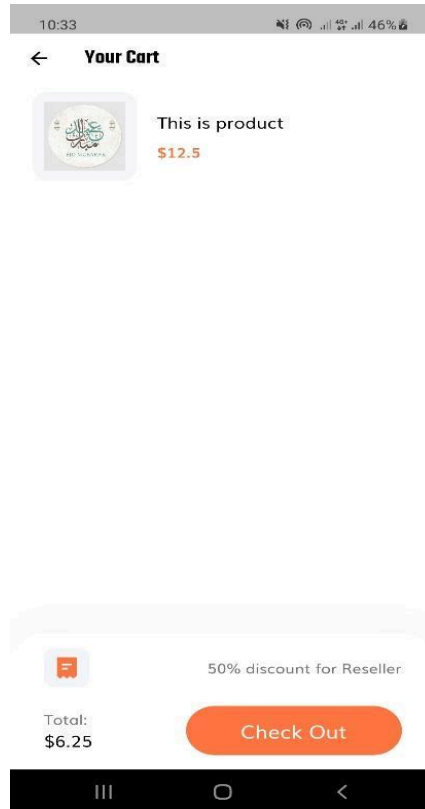


Figure 39: view cart Add Products to the Cart

5.3 System Schedule & Design

5.3.1 Activity Schedule

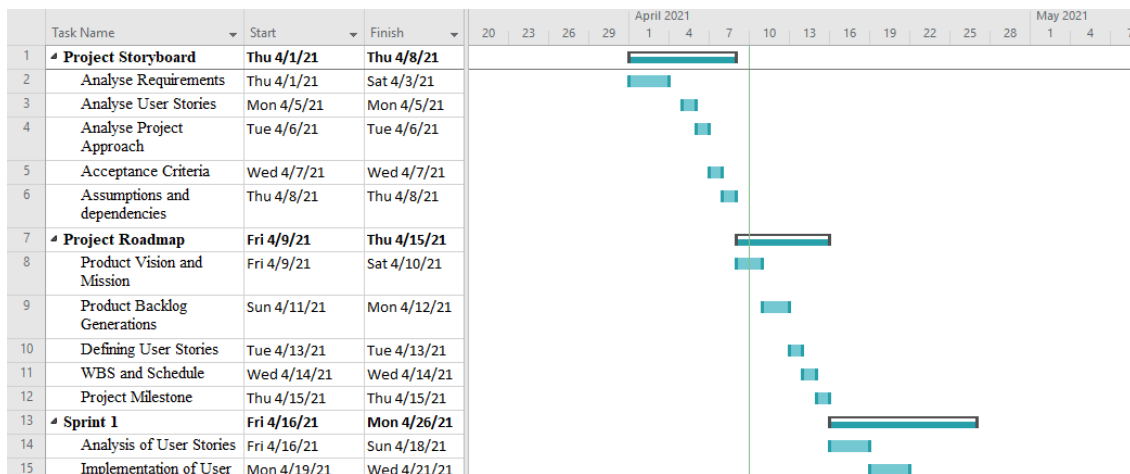


Figure 40: Gantt Chart No1

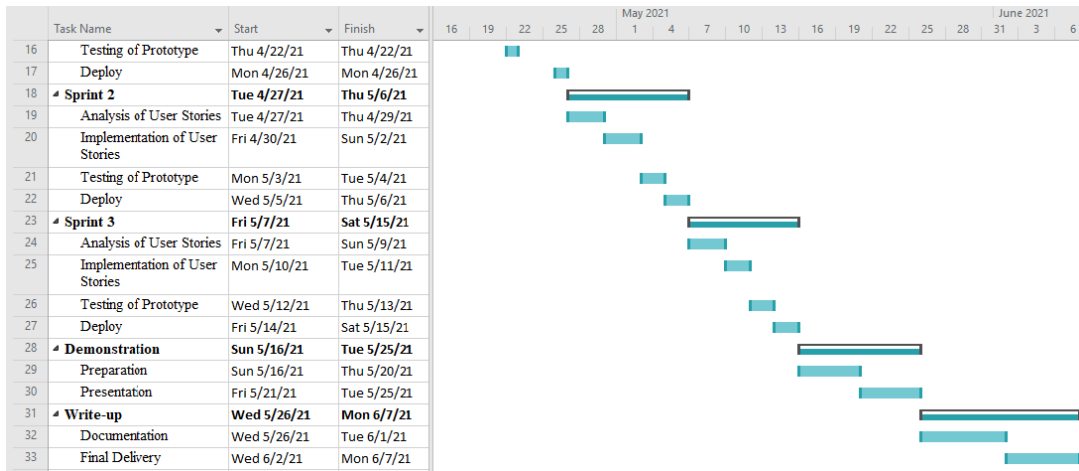


Figure 41: Figure 42: Gantt Chart 2

5.3.2 Work Breakdown Structure of Project

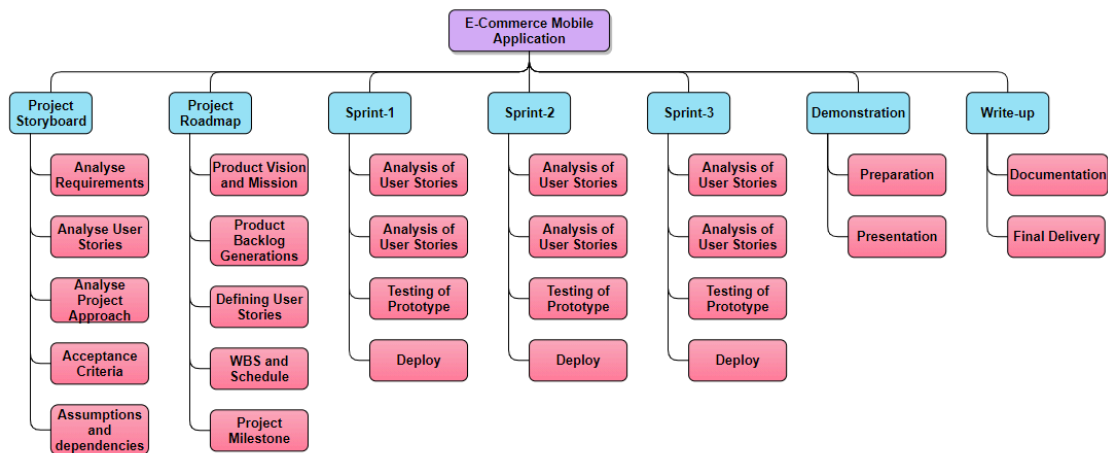


Figure 43: WBS of Project

5.3.3 Use Case

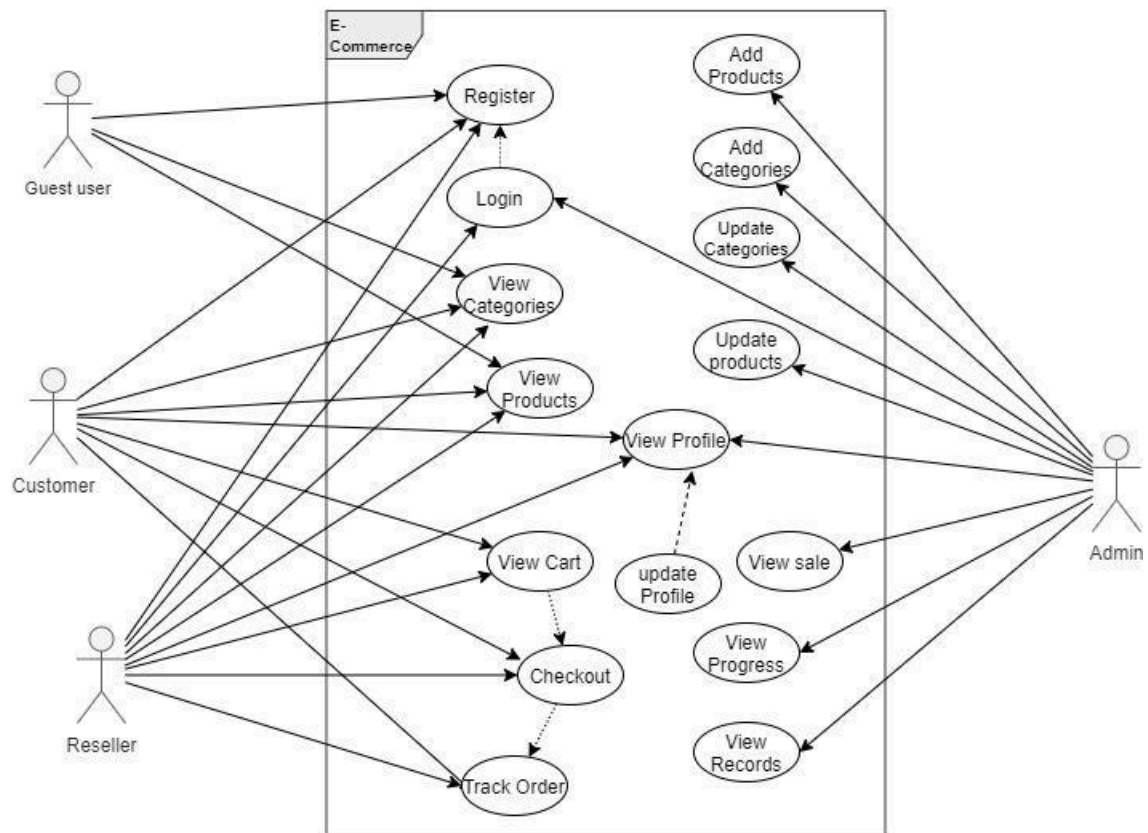


Figure 44: Use Case Diagram

5.3.4 Event Scenario or Sequence

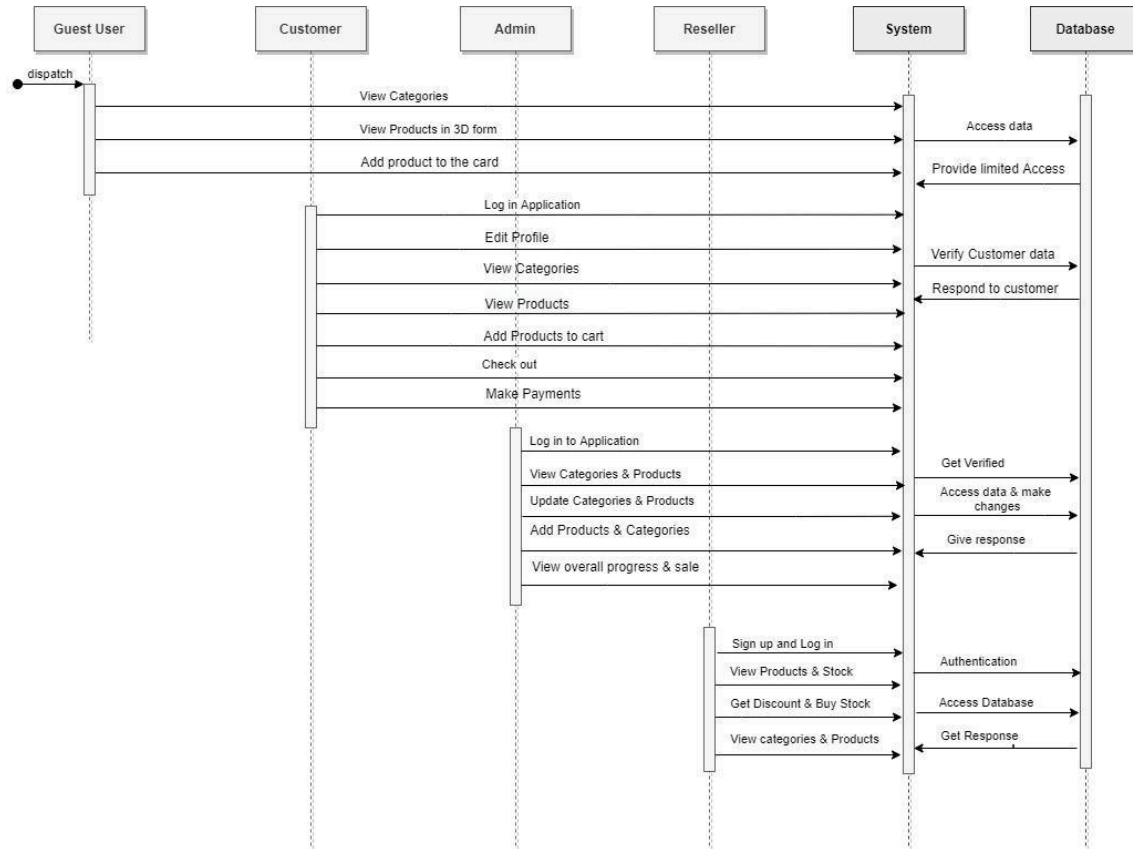


Figure 45: Sequence Diagram

5.3.5 Entities Relationships

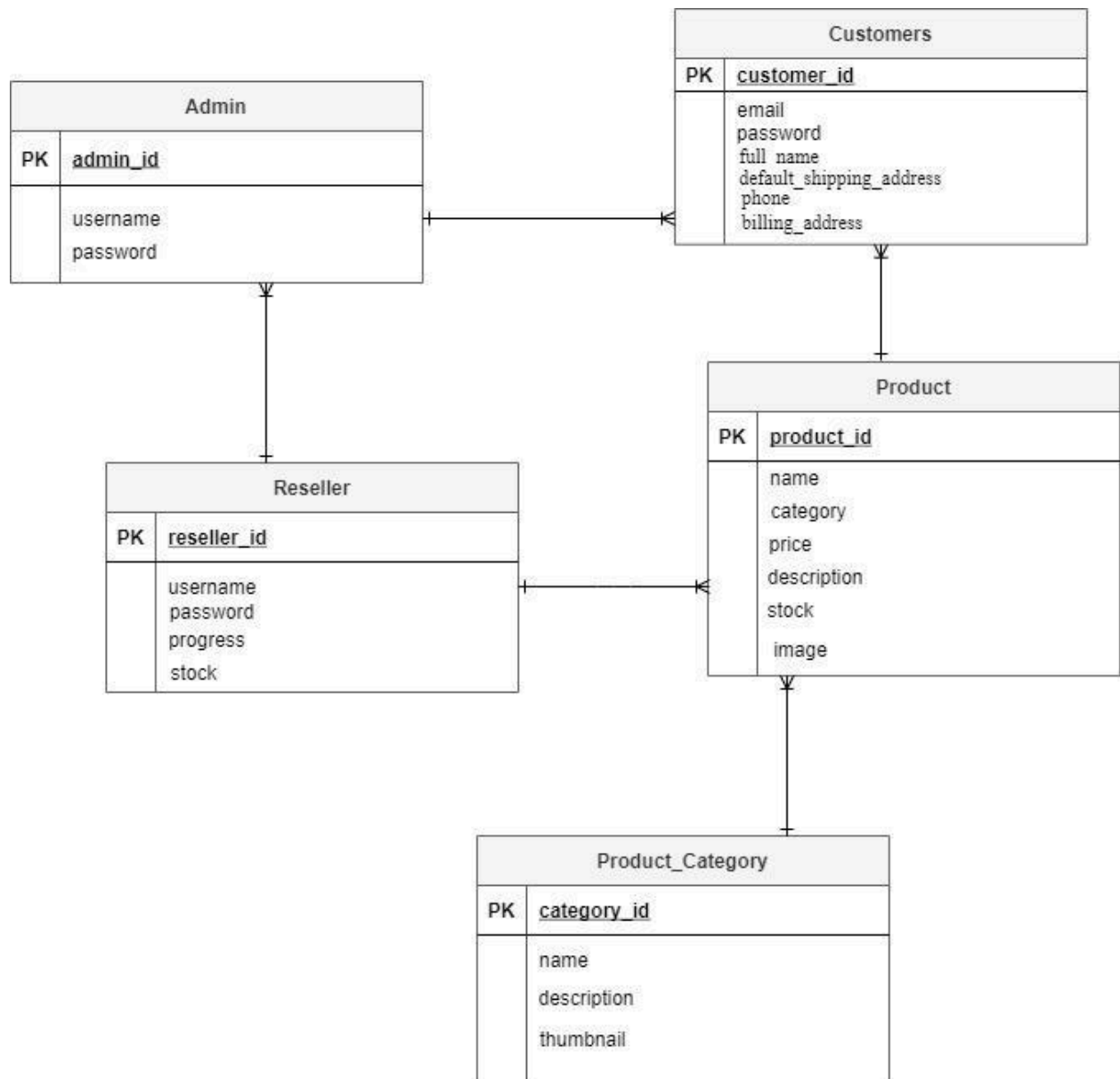


Figure 46: ER Diagram

5.3.6 Business Process Model

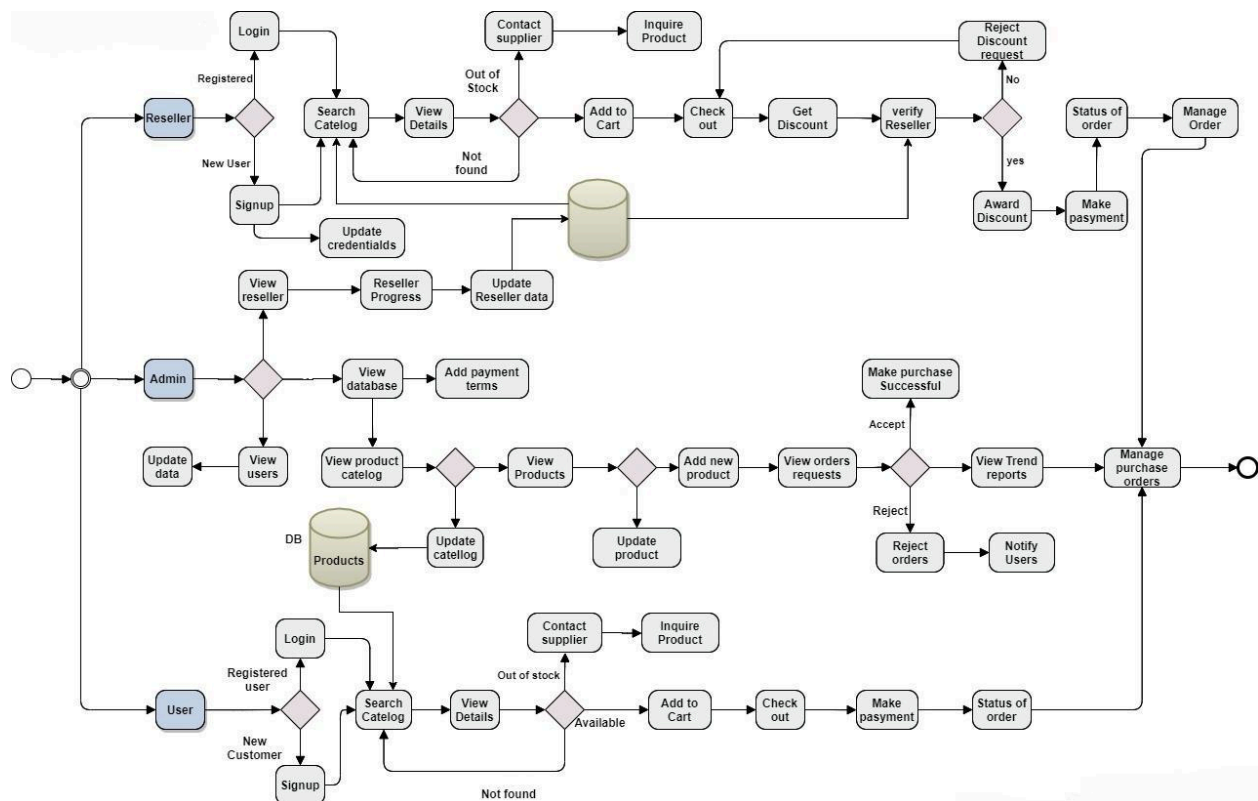


Figure 47: BPMN

5.4 Testing

5.4.1 Testing Strategies

Software testing is as old as the hills of digital computer history. Software testing is one Important resource for assessing the quality of software. Because testing is commonly used 40 – 50% of development efforts and spends more labour on systems that require a higher level of Reliability, it is an important part of software engineering. For testing our product, the agile testing system has been used. As a consistent method to manage steady procedure has been used for advancement, a slow method to manage testing has been finished. The waterfall procedure contains methods of checking to grow the thing's consistency. The assessment in the agile methodology is introduced without the completion of the task because it supports iteration of activities. As the features are introduced in unquestionable runs, the testing begins early and is done constantly. The analyser performed testing on the other hand the developer write code in the testing method. The goal of testing as customer stories wants as examinations, nature of what

ought to be surveyed and methods of how tests can be coordinated are associated with the agile testing framework (Luo, n.d.).

5.4.2 Types of Testing

Testing types are categorized into two main categories at a broader level. These two terms are functional testing and non-functional testing. Both are used according to their criteria of testing.

5.4.2.1 Non-functional Testing

It is performed to test the non-functional facts of the software. These non-functional aspects are performance, integrity, security, usability, productivity, and reliability etc. It is done to improve software quality. With this type of testing, we could make our software better.

5.4.2.2 Functional Testing

This testing includes the testing of the useful parts of a product application. When we are testing the application then we must test every single functionality. At the end of testing, we must observe we are getting the required outcomes or not. The types of this testing we have used for our project are:

- **Unit Testing**

In this testing, every single functional part of the project is tested so that complications are removed before combining each part as a system.

- **Integration Testing**

After unit testing successfully all the parts were combined to make a product. This integrated part was tested as a whole, and results were compiled.

- **Interface Testing**

It is as important as other testing methodologies are. It was done to check the interfaces are working properly or not. We checked each interface one by one to find flaws in the working of the user interfaces.

- **Security Testing**

Security and integrity is the most vital part of software development. If the software is accessing the user personal data, then there must be the integrity and security of data ensured before making them part of the database. The security level of each module is tested completely for ensuring the customer that data will not be accessed by an unauthorized person.

- **System Testing**

System testing is done to make sure the whole system is working according to the requirements specified by the client. Each part in this testing is analysed and tested. It is the more reliable form of testing which ensure the productivity of the product.



Figure 48: Types of Functional Testing

5.4.2 Test Cases

Table 2: Test Cases of Project

ID	Test Description	Pre-condition	Testing Condition	Expect Outcome	Act Outcome	Pass/Fail
TC-001	Test the login page of the application of all the admin panels.	The admin logs in to the application.	The login page is displayed to the admin.	Successfull y log in to the application.	Successfull y log in to the application.	Pass

TC-00 2	Test the sign-up page of the admin panel of the application.	The admin signs up in the application.	The sign-up page is displayed to every user panel.	Successful sign up in the application.	Successful sign up in the application.	Pass
TC-00 3	Test the add category page of the application.	The added category in the database.	The add category page is displayed to the admin.	Successful add a category.	Successful add a category.	Pass
TC-00 4	Test the update category page of the application.	The admin update category in the application.	The update category page is displayed to the admin.	Successful update category	Successful update category	Pass
TC-00 5	Test the updated reseller page of the application.	The admin update resellers in the application	The update reseller page is displayed to the admin.	Successful update reseller in the application.	Successful update reseller in the application.	Pass
TC-00 6	Test the add product page to the application.	The admin add the product to the application	The add product page is displayed to the admin.	Successfully add product in the application.	Successfully add product in the application.	Pass
TC-00 7	Test the update product page of the application.	The admin update product in the application	The updated product page is displayed to the admin.	Successful update the product in the application.	Successful update the product in the application.	Pass

TC-00 6	Test the login page of the application for the reseller.	The reseller logs in to the application.	The login page is displayed to the reseller.	Successful y log in to the application.	Successful y log in to the application.	Pass
TC-00 7	Test the sign-up page of the reseller application.	The reseller signs up in the application.	The sign-up page is displayed to the reseller.	Successful y sign up in the application.	Successful y sign up in the application.	Pass
TC-00 8	Test the view category of the application.	The reseller view categories of products.	The view category page is displayed to the reseller.	Successful y view categories.	Successful y view categories.	Pass
TC-00 9	Test the view product of the application.	The reseller views all the products.	The view product page is displayed to the reseller.	Successful y view products.	Successful y view products.	Pass
TC-01 0	Test the view detail page for every product.	The reseller views the detail of the products.	The page for product detail is shown for the reseller.	Successful y shoe detail of the product.	Successful y shoe detail of product	Pass
TC-01 1	Test checkout page for reseller.	The reseller checkouts for purchasing products.	The checkout page is shown.	Successful y checkout for any product.	Successful y checkout for any product.	Pass
TC-01 2	Test the track order page of the application for the reseller.	The reseller tracks his order.	The track order page is shown properly.	Successful y tracked the	Successful y tracked the	Pass

				concerned order.	concerned order.	
TC-01 3	Test update the profile page for reseller.	The reseller updates his profile for changing personal details.	The update profile page is properly shown.	Successfull y updated profile.	Successfull y updated profile.	Pass
TC-01 4	Test the login page of the application for the customer.	The customer logs in to the application.	The login page is displayed to the customer.	Successfull y log in to the application.	Successfull y log in to the application.	Pass
TC-01 5	Test the sign-up page of the customer application.	The customer signs up in the application.	The sign-up page is displayed to the customer.	Successfull y sign up in the application.	Successfull y sign up in the application.	Pass
TC-01 6	Test the view category of the application.	The customer view categories of products.	The view category page is displayed to the customer.	Successfull y view categories.	Successfull y view categories.	Pass
TC-01 7	Test the view product of the application.	The customer views all the products.	The view product page is displayed to the customer.	Successfull y view products.	Successfull y view products.	Pass
TC-01 8	Test the view detail page for every product.	The customer views the detail of the products.	The page for product detail is shown for the reseller.	Successfull y shoe detail of the product.	Successfull y show the detail of the product	Pass

TC-019	Test view order details for the customer.	The customer view order detail.	The track order page is shown properly.	Successfully tracked the concerned order.	Successfully tracked the concerned order.	Pass
TC-020	Test update profile page for customer.	The customer updates his profile for changing personal details.	The update profile page is properly shown.	Successfully updated profile.	Successfully updated profile.	Pass
TC-021	Test the view product in 3D form page.,	The guest user can view products in 3D form.	The page is properly shown.	Successfully updated profile.	Successfully updated profile.	Pass

Chapter No 6

SCOPE, LIMITATIONS AND RECOMMENDATIONS

6.1 Limitations

Following are the limitations of the project:

- Guest user cannot log in to the application.
- Guest user cannot check out for purchasing product.
- Guest user cannot view any progress.
- The customer cannot view the reseller progress.
- Customer cannot get discount as compared to the reseller.
- Admin cannot track the activity of the guest user.
- The application has no facility of chatbot where the reseller and user contact the admin.
- The application has no facility for feedback from the customer.

- The application has no facility for adding products to the wish list.
- The application has no feature of making monthly sale reports automatically.

6.2 Recommendations

Following is the recommendation for this project:

- Guest user must have a popup that suggests he register himself.
- Guest user must contact the admin for making queries.
- The guest user should have a view of the progress report to increase his trust in the supplier.
- Customer should be able to give feedback after receiving products.
- Customer should be able to view reviews of other customers on the products.
- Customer should be able to add products to wish list.
- The application must have a feature of discounts for customers on special days.
- The application should have a feature of a chatbot.
- The application must provide the facility of FAQs for customers.
- The application should generate a monthly progress chart for users it can increase the trust in the application.
- The application must have advanced features of the graphical interface.

6.3 Future Scope

The future scope can be determined from the following key points:

- This application will be inseparable from the web because the use of apps for e-shopping becoming popular gradually.
- This application will have a big part in the growth of sale as well as promotions.
- It will make the platform of business activities and shopping broad.
- It will provide a place where business set-up cost would be minimal.
- It will increase the attention of international customers.

Chapter No 7

CONCLUSION

7.1 Conclusion

This application has been developed for facilitating customers in their e-shopping related activities. This application is developed for increasing the demand for virtual shopping pints that can serve internationally. It is a place to serve people with easy grab product and services. The application facilitates the reseller with discounted products. It is like the wholesale rate. This application has been increased the range of the products for the customer from which they can easily choose the required one. This encourages the customers for more shopping activities in future.

7.2 Client's Feedback

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Appendix

9.1 Backlog of Sprint-1

Table 3: Sprint-1 Backlog

US-No	Stories	Satisfactory Conditions	Estimated Tasks
US-01	As an admin, I want to manage the whole application so that I can handle the application.	Admin can use the main page to manage all the activities of his panel.	T1: Design the main page for the application. 3 Hours---Aqib Syed Mustafa Ali
US-02	As an admin, I want to log in so that I can manage the application.	The application must have a login feature for the application. Admin can open the login page and log into the app using his credentials.	T2: Design Page for login of admin in the application. 2 Hours—Aqib Syed Mustafa Ali T3: Attach the login page to the application. 1 Hour---Mehmood Ali T4: Create a database to store the information. 3 Hours---Mohammed Vazeer Ahmed T5: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-03	As an admin, I want to edit my profile to make changes to my details.	There must be a feature for admin used to edit admin profile.	T6: Design Page for editing admin profile.

		Admin must be able to edit his profile, e.g., change of password, etc.	2 Hours---Aqib Syed Mustafa Ali T7: Attach the editing page to the application. 1 Hour--- Mehmood Ali T8: Create a Database to store admin credentials. 2 Hours--- Mohammed Vazeer Ahmed T9: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-004	As an admin, I want to view products so that I can view all the available products.	The application should contain a feature from where he can see all the products. Admin must be able to view all the products.	T10: Design Page that displays all the products. 2 Hours---Aqib Syed Mustafa Ali T11: Attach the page to the main menu. 1 Hour--- Mehmood Ali T12: Test the working of the page. 1 Hour---Nizam Uddin Mohammed

US-05	As an admin, I want to add products so that I can provide new products to users.	There must be a feature in the application that makes the admin able to add products. Admin must be able to add products to the application.	T13: Design Page for adding products to the Database. 1 Hour---Aqib Syed Mustafa Ali T14: Attach the page to the main menu. 1 Hour--- Mehmood Ali T15: Create a Database to store all the new products. 2 Hour---Mohammed Vazeer Ahmed T16: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-06	As an admin, I want to modify products so that I can update product detail.	The application must have a feature for the modification of products. Admin must be able to modify all the existing products.	T17: Design Page for modifying products. 2 Hour---Aqib Syed Mustafa Ali T18: Attach the page to the main menu. 1 Hour--- Mehmood Ali T19: Create a Database to store all updated information of products.

			2 Hour--- Mohammed Vazeer Ahmed T20: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-007	As an admin, I want to view categories so that I can view available categories for products.	The application must have a feature to view all the product categories. Admin must be able to view all the available categories of the products.	T21: Design Page that displays all the available categories of the product. 2 Hours---Aqib Syed Mustafa Ali T22: Attach the page to the main menu. 1 Hour ---Mehmood Ali T23: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-008	As an admin, I want to modify categories so that I can update product categories.	The application must contain a feature for modification of the product categories. Admin must be able to modify categories.	T24: Design module used to modify product categories. 3 Hours---Aqib Syed Mustafa Ali T25: Attach the page to the main menu. 1 Hour--- Mehmood Ali

			T26: Create a Database to store all the modification details. 4 Hours--- Mohammed Vazeer Ahmed T27: Test the working of the page. 1 Hour ---Nizam Uddin Mohammed
US-009	As an admin, I want to view the overall sales of products so that I can analyze how much profit the business gets.	There must be a feature for displaying all the sales of the products. Admin must be able to view overall sales.	T28: Design Page for displaying overall sales. 2 Hours---Aqib Syed Mustafa Ali T29: Attach the page to the application. 1 Hour--- Mehmood Ali T30: Create a Database to store all sale details. 5 Hours---Mohammed Vazeer Ahmed T31: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-010	As an admin, I want to view user profiles so	The application should contain a feature for viewing profiles of users.	T32: Design Page for viewing user profiles.

	that I can view all the users of the application.	Admin must be able to view the user's profile.	2 Hours---Aqib Syed Mustafa Ali T33: Attach the page to the application. 2 Hours---Mehmood Ali T34: Test the working of the page. 2 Hours---Nizam Uddin Mohammed
US-011	As an admin, I want to edit users of the application so that I can update the details of application users.	The application must have a feature that allows editing of the current users. The admin should be able to modify the current users of the application.	T35: Design Page for editing user profiles. 1 Hour---Aqib Syed Mustafa Ali T36: Attach the page to the application. 1 Hour--- Mehmood Ali T37: Create a Database that allows the admin to edit users. 2 Hours--- Mohammed Vazeer Ahmed T38: Test the working of the page. 1 Hour---Nizam Uddin Mohammed

US-012	As an admin, I want to view resellers' progress so that I can analyze how many sales they made.	There must be a feature that allows the admin to view the progress of resellers. Admin must be able to view the overall progress of resellers.	T39: Design Page that allows admin to view reseller progress. 4 Hours---Aqib Syed Mustafa Ali T40: Attach the page to the application. 1 Hour---Mehmood Ali T41: Create a Database to store all reseller progress. 3 Hours---Farooq T42: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-013	As an admin, I want to view and edit profiles of resellers so that I can view and update their details.	There must be a feature for editing profiles of resellers. Admin must be able to update details of the resellers.	T43: Design Page that allows admin to edit reseller profile. 3 hours--- Aqib Syed Mustafa Ali T44: Attach the page to the application. 2 Hours--- Mehmood Ali T45: Create a Database to store all the changes for resellers. 4 Hours---Mohammed Vazeer Ahmed

			T46: Test the working of the page. 2 Hours---Nizam Uddin Mohammed
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9.2 Backlog of Sprint-2

Table 4: Sprint-2 Backlog

US-No	Stories	Satisfactory Conditions	Estimated Tasks
US-014	AS a reseller, I want to register my account so that I can use the application.	The application must have a sign-up feature for reseller. The reseller can use the signup page to register his account.	T1: Design Page for login of admin in the application. T2: Attach the sign-up page to the application. 2 Hour---Mehmood Ali T3: Create a database to store the information. 2 Hours---Mohammed Vazeer Ahmed T4: Test the working of the page. 2 Hour---Nizam Uddin Mohammed
US-015	As a reseller, I want to log in so that I can	The application must have a login feature for the application.	T5: Design Page for reseller's login in the application.

	view all the products and my purchases.	The reseller can open the login page and log into the app using his credentials.	2 Hours—Aqib Syed Mustafa Ali T6: Attach the login page to the application. 1 Hour---Mehmood Ali T7: Create a database to store the information. 3 Hours---Mohammed Vazeer Ahmed T8: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-016	As a reseller, I want to view all the available categories so that I can select the suitable one for my purchases.	There must be a page for reseller used to view all the available categories. The reseller must be able to open the page and view the categories.	T9: Design Page for the reseller to view all the categories 2 Hours---Aqib Syed Mustafa Ali T10: Attach the categories page to the application. 1 Hour--- Mehmood Ali T11: Create a Database to access the required data. 2 Hours--- Mohammed Vazeer Ahmed

			T12: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-017	As a reseller, I want to add to the cart all the products so that I can initiate the process of purchasing.	There must be a feature in the application for the reseller to add products to his cart. The reseller must be able to add products to the cart.	T13: Design a Page that allows the reseller to add products to the cart. 2 Hours---Aqib Syed Mustafa Ali T14: Attach the page to the main menu. 1 Hour--- Mehmood Ali T15: Create a Database to store all the new products. T16: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-018	As a reseller, I want to view the stock of the product so that I can purchase the products.	There must be a feature in the application for the reseller to view his stock. The reseller must be able to open the page and view his stock.	T17: Design Page for the reseller to view all the products. 1 Hour---Aqib Syed Mustafa Ali T18: Attach the page to the main menu. 1 Hour--- Mehmood Ali

			T19: Create a database to access the products. 2 Hour---Mohammed Vazeer Ahmed T20: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-019	As a reseller, I must be able to get a discount.	There must be a feature in the application that makes the reseller able to get a discount on products. Reseller must be able to get discount on the products.	T21: Design a discount page. 2 Hour---Aqib Syed Mustafa Ali T22: Attach the page to the main menu. 1 Hour--- Mehmoood Ali T23: Create a Database to store all information. 2 Hour--- Mohammed Vazeer Ahmed T24: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-020	As a reseller, I want to purchase products so that I can use	The application must have a feature for the reseller to purchase the products for his stock.	T25: Design Page for adding products to the cart for purchase.

	them for my reselling.	Reseller must be able to purchase products for his stock.	2 Hours---Aqib Syed Mustafa Ali T26: Attach the page to the main menu. 1 Hour ---Mehmood Ali T27: Create a Database to store all the information related to the purchase. T28: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-021	As a reseller, I want to check the order detail so that I track back my orders.	There must be a feature for reseller for checking order details. The reseller must be able to check order details.	T29: Design module used to check order details. 3 Hours---Aqib Syed Mustafa Ali T30: Attach the page to the main menu. 1 Hour--- Mehmood Ali T31: Create a Database to access all the details. 4 Hours--- Mohammed Vazeer Ahmed T32: Test the working of the page.

			1 Hour ---Nizam Uddin Mohammed
US-0 22	As a reseller, I want to view products details so that I can select the best-suited products for me.	There must be a feature for checking products details. Reseller must be able to view products detail.	T33: Design Page for displaying specific products with related detail. 2 Hours---Aqib Syed Mustafa Ali T34: Attach the page to the application. 1 Hour--- Mehmood Ali T35: Create a Database for accessing required products. 5 Hours---Mohammed Vazeer Ahmed T36: Test the working of the page. 1 Hour---Nizam Uddin Mohammed

9.3 Backlog of Sprint-3

Table 5: Sprint-3 Backlog

US- No	Stories	Satisfactory Conditions	Estimated Tasks
US-023	As a customer, I want to log in with an email	The application must have a sign-in page.	T1: Design Page for login of admin in the application.

	and password so that I can use the application.	The customer can sign in to the application.	2 Hours—Aqib Syed Mustafa Ali T2: Attach the log-in page to the application. 1 Hour---Mehmood Ali T3: Create a database to store the information. 2 Hours---Mohammed Vazeer Ahmed T4: Test the working of the page. 2 Hour---Nizam Uddin Mohammed
US-024	As a customer, I want to view my profile so that I can check my details.	The application must have a view profile page for the customer. The customer can view his detail on the view profile page.	T5: Design Page for view profile for guest users. 2 Hours—Aqib Syed Mustafa Ali T6: Attach page to the application. 2 Hour---Mehmood Ali T7: Create a database to access the information. 3 Hours---Mohammed Vazeer Ahmed T8: Test the working of the page.

			2 Hour---Nizam Uddin Mohammed
US-025	As a customer, I want to view the profile and change the password so that I can update my profile.	The application must have the feature of updating its profile. The customer can be able to update his profile.	T9: Design Page for the customer to update his profile. 2 Hours---Aqib Syed Mustafa Ali T10: Attach the page to the application. 1 Hour--- Mehmood Ali T11: Create a Database to store the updated data. 1 Hour--- Mohammed Vazeer Ahmed T12: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-026	As a customer, I want to view order details so that I can check the progress of product shipment.	There must be a page for tracking specific order for the customer. The customer must be able to track his order after shipment.	T13: Design a Page for tracking the order. 2 Hours---Aqib Syed Mustafa Ali T14: Attach the page to the main menu. 1 Hour--- Mehmood Ali

			T15: Create a Database to access all stored data related to the purchased products. T16: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-027	As a customer, I want to check my wallet balance so that I can make money transactions.	There must be a feature for checking the wallet balance of the customer. The customer should be able to check his wallet balance.	T17: Design Page for the checking wallet balance. 1 Hour---Aqib Syed Mustafa Ali T18: Attach the page to the main menu. 1 Hour--- Mehmoood Ali T19: Create a database to access the record. 2 Hour---Mohammed Vazeer Ahmed T20: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-028	As a guest user, I want to view the available categories so that I can select the required product.	There must be a feature for the guest user to view all the available categories of the products.	T21: Design a page for viewing categories. 2 Hour---Aqib Syed Mustafa Ali T22: Attach the page to the main menu.

		Guest user must be able to view all the categories.	1 Hour--- Mehmood Ali T23: Create a Database to access all records. 2 Hour--- Mohammed Vazeer Ahmed T24: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-029	As a guest user, I want to view the products so that I can read all the detail.	There must be a feature for the guest user to view the products. Guest user must be able to view the products.	T25: Design Page for viewing the products. 2 Hours---Aqib Syed Mustafa Ali T26: Attach the page to the main menu. 1 Hour ---Mehmood Ali T27: Create a Database to access all the information. T28: Test the working of the page. 1 Hour---Nizam Uddin Mohammed
US-030	As a guest user, I want to view the products in 3D form so that I can	There must be a feature for a guest user to view products in 3D form.	T29: Design module used to check order details. 3 Hours---Aqib Syed Mustafa Ali

	check product specifications.	The guest user must be able to view products in 3D form.	T30: Attach the page to the main menu. 1 Hour--- Mehmood Ali T31: Create a Database to access all the details. 4 Hours--- Mohammed Vazeer Ahmed T32: Test the working of the page. 1 Hour ---Nizam Uddin Mohammed
US-031	As a guest user, I want to view product details so that I can find the right one.	There must be a feature for checking products details. Guest user must be able to view products detail.	T33: Design Page for displaying specific products with related detail. 2 Hours---Aqib Syed Mustafa Ali T34: Attach the page to the application. 1 Hour--- Mehmood Ali T35: Create a Database for accessing required products. 3 Hours---Mohammed Vazeer Ahmed T36: Test the working of the page.

Problem Statement <i>Is the project problem clearly explained? Are the relevant stakeholders correctly identified?</i> <i>Are user stories specified with acceptance criteria included?</i>	2 2	
Project Objectives <i>Is the aim of the project clearly stated to align with a solution to the project problem?</i>	2	
Related Academic and Commercial Research <i>Is a structured review of relevant literature presented?</i> <i>What conclusions are drawn from the review of relevant literature?</i> <i>How does the literature inform the proposed product as a solution to the project problem?</i> <i>What relevant theories, perspectives, or frameworks have been adopted to address the project problem? Is the Justification provided for the selected theory/framework?</i>	3 2 3 3	
Project Approach <i>Explanation of the project approach, its application in the project, and specific tasks undertaken in the product development.</i>	5	
Discussion of Product(s) <i>Comprehensive description/demonstration of the product developed</i> <i>Results and findings from product testing/evaluation as per acceptance criteria</i>	5 5	
Summary, Limitations, and Recommendation <i>What are the applications of the product?</i> <i>What are the limitations of the product?</i> <i>How might the product be improved?</i>	2 2 2	
References <i>Are references relevant?</i> <i>Are they correctly cited and listed?</i>	2 2	
Total	45	
Comments & Feedback :		